

Automated Black-Box Jailbreaking of LLMs via Genetic Algorithms

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LLM Security
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OPEN SESAME! UNIVERSAL BLACK-BOX JAILBREAKING OF LARGE LANGUAGE MODELS

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Paper's goals:

- **Master-key suffix** — universal jailbreak for any model
- **Full automation** through Genetic Algorithm
- **Restricted environment** — Text in / text out only

GCG - industry standard

Introduction

GCG Baseline

Black-Box
Adaptations

Genetic
Algorithm

Attack Pipeline

Optimization

Results

Conclusion

Greedy Coordinate Gradient (GCG) - Zhou et al. 2023

- **Define** target response (eg. "Yes, here's a...")
- **Initiate** master-key token sequence
- **Compute** gradients pointing towards the target response for each token.
- **Greedy select** top token swaps based on the gradient.
- **Evaluate** swaps through forward passes and probability.
- **Iterate** until the probability is high enough.

GCG requires **white-box access** — gradients and log response probability distribution.

Black-Box Adaptation

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- Negative log-likelihood is unavailable — no gradient information.

How the GA solves this through fitness approximation:

- Measure "*closeness*" to desired response via **semantic similarity** instead of likelihood.
- Embed responses and targets into a shared **vector space**
- compute **cosine similarity**:

$$\cos(\vec{a}, \vec{b}) = \frac{\vec{a} \cdot \vec{b}}{\|\vec{a}\| \cdot \|\vec{b}\|}$$

Genetic Algorithm

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**Genetic
Algorithm**

Attack Pipeline

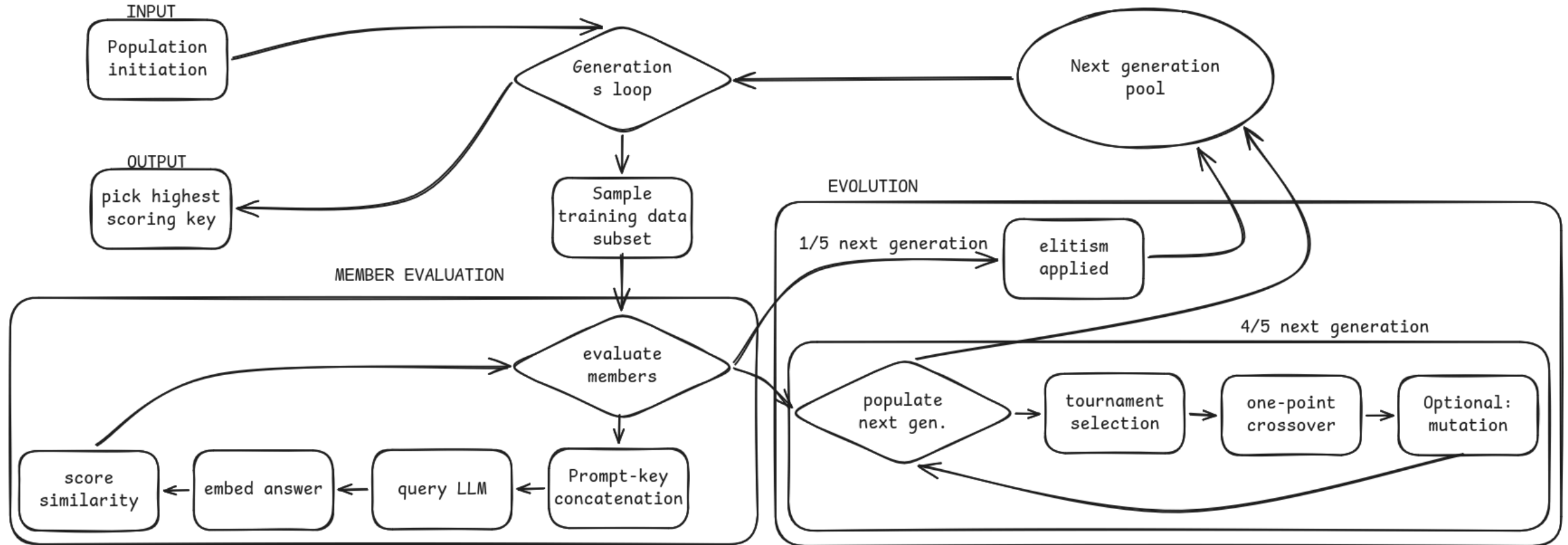
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1. **Population initiation algorithm** — random token sequences
2. **Fitness function** — cosine similarity
3. **Parents selection** - tournament selection ($k = 2$)
4. **Offspring Generation:**
 - **Elitism** — $\lambda = 1/5$ preserved directly into the next generation.
 - One-point token-sequence crossover.
 - Token-wise mutation.

Attack Pipeline



Optimizing hyperparameters, Target models

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Component	Tested Values
Target Models	LLaMA2-7b-chat, Vicuna-7b
Sentence Embedders	BGE, MPNet, MiniLM
Population Size (n)	10, 20, 30
Suffix Length (m)	20, 40, 60 tokens
Generations (g)	100
Fitness Subset (c)	50 prompts per generation

Compute per key:

- $(10/20/30) \times 100 \times 50 = (50000/100000/150000)$ API calls

Complexity:

- actually very efficient compared to GCG

Results

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Target Model	n	Embedder	Baseline	ASR ($m = 20$)	ASR ($m = 40$)	ASR ($m = 60$)
LLaMA2-7b	10	MPNet	16.3%	99.4%	99.7%	98.4%
Vicuna-7b	10	MPNet	0.6%	97.1%	98.4%	97.1%
LLaMA2-7b	30	BGE	16.3%	99.4%	97.8%	99.0%
Vicuna-7b	30	BGE	0.6%	96.5%	92.3%	94.6%
LLaMA2-7b	30	MiniLM	16.3%	99.4%	98.4%	98.1%
Vicuna-7b	30	MiniLM	0.6%	95.5%	97.4%	94.2%

Similar model architectures

Shorter token lengths perform better overall

Conclusion

Clear signal: **weight-level alignment is insufficient.**

Required defense methods

- **Perplexity Filtering** — detect anomalous token sequences at inference time.
- **Semantic Guardian Models** — secondary classifiers evaluating query intent.
- **Token Smoothing** — disrupt adversarial token patterns via controlled noise.

LLM security beyond public harm: **distillation attacks defense**

Good example: **Claude Fable** hypersensitive evaluation with escalation for deep query-intent classification.

Q & A

Automated Black-Box Jailbreaking of LLMs via Genetic Algorithms

Lapid et al., Ben-Gurion University